Unit 5, Lesson 10: Converting Units Using Ratios

Benchmark

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.

Learning Target

- ☐ I can use ratios to convert units within the same measurement system.

5.10.1 Warm-Up: Furniture Designer

1. Jared Rusten is a woodworker who creates modern furniture in San Francisco.

    a. Jared usually starts designing furniture by sketching his ideas before he makes a scaled drawing using measurements to build the actual piece of furniture. What do you think a scale drawing is?

    b. Jared said, “I never thought that I’d be using math as much as I do now.” What are some of the ways he uses math as a furniture designer and maker?

5.10.2 Guided Instruction: Converting Units of Measure Using Ratio Relationships

Many different units are used to measure lengths, weight, capacity (volume), and time.

1. What are some units of measure that are used to take each type of measurement?

Length	Weight	Capacity	Time

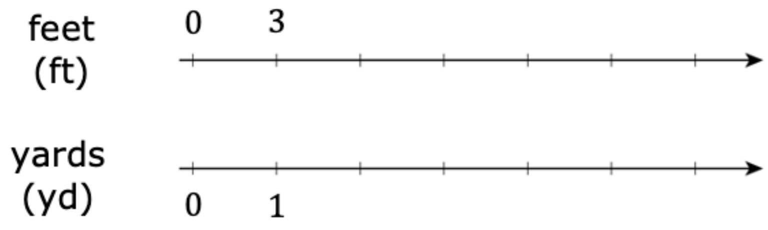
The United States uses the customary system of measurement. The table shows the unit conversions for units within this system.

2. Label each row using the words **Length, Weight,** and **Capacity** to indicate what each set of units measures.

Type of Measurement	Unit Conversions in the Customary System
	1 foot (ft) = 12 inches (in.) 1 yard (yd) = 3 feet (ft) 1 mile (mi) = 5,280 feet (ft) 1 mile (mi) = 1,760 yards (yd)
	1 cup (c) = 8 fluid ounces (fl oz) 1 pint (pt) = 2 cups (c) 1 quart (qt) = 2 pints (pt) 1 gallon (gal.) = 4 quarts (qt)
	1 pound (lb) = 16 ounces (oz) 1 ton (t) = 2,000 pounds (lb)

Each conversion rate represents a ratio. Double number line diagrams can be used to convert units.

3. Use the double number line diagram shown to find how many feet (ft) are in 4 yards (yd).



There are _____ feet in 4 yards.

Tables can also be used to convert units of measure.

4. The average weight of an African forest elephant is $6{,}000$ pounds. Use the unit conversion and the table to determine how many tons the average African forest elephant weighs.

Pounds	Tons
2,000	1

Notice that each conversion rate is a unit rate.

Unit rates can be used as factors to convert from one unit to another.

For example, the factor $\frac{4 \text{ quarts}}{1 \text{ gallon}}$ is equivalent to 1 because the numerator and denominator are equivalent values.

Multiplying by factors equivalent to 1 will result in equivalent values.

Kaluba determined how many quarts are in 2 gallons of milk using this strategy. His work is shown.

$$\frac{2 \text{ gal.}}{1} \times \frac{4 \text{ qt}}{1 \text{ gal.}} = 8 \text{ qt}$$

Kaluba explained, "In my calculation, the unit 'gallons' is in the numerator and the denominator. This means I can eliminate this unit because it becomes 1, similar to when equivalent values are in the numerator and denominator. The result of the conversion is then the number of quarts."

5. Complete the calculation to find how many fluid ounces (fl oz) are in 3 pints (pt).

$$\frac{3 \text{ pt}}{1} \times \frac{\quad\quad}{1 \text{ pt}} \times \frac{\quad\quad \text{fl oz}}{\quad\quad} = \quad\quad \text{fl oz}$$

The units of measure used in the United States are ***customary units***.

The units of measure used in most of the world are ***metric units***. Like the decimal system, the metric system uses the base 10.

The tables below show additional unit conversion rates for other units of measure.

Type of Measurement	Unit Conversions in the Metric System
Length	1 meter (m) = 100 centimeters (cm) 1 meter (m) = 1000 millimeters (mm) 1 kilometer (km) = 1000 meters (m)
Capacity	1 liter (L) = 1000 milliliters (mL)
Weight or Mass	1 gram (g) = 1000 milligrams (mg) 1 kilogram (kg) = 1000 grams (g)

Time Conversions
1 minute (min) = 60 seconds (s) 1 hour (h) = 60 minutes (min) 1 day = 24 hours (h) 1 year = 365 days 1 year = 52 weeks 1 week = 7 days

5.10.3 Guided Practice: Converting Units of Measure Using Unit Rates

1. Use the conversion rates to find each value. Show your work.

   a. 2.62 km = _____ m

   b. 9 in. = _____ yd

   c. 1 week = _____ hours

2. Adrian weighed 7 pounds (lb) and 4 ounces (oz) when he was born. How many ounces did Adrian weigh when he was born?

3. Ibuprofen is a medicine sometimes used for pain relief, such as for a headache. For adults, it is not safe to take more than 3200 milligrams (mg) of ibuprofen each day. How many grams of ibuprofen is safe for adults to take in one day?

5.10.4 Collaboration: Error Analysis

1. Valerie makes sure she drinks 8 cups (c) of water each day. She wondered how many gallons of water this is, so she used conversion rates to find out. Her work is shown.

$$\frac{80\text{ c}}{1} \times \frac{2}{1} \times \frac{1}{2} \times \frac{1}{4} = \frac{160}{8} = 20\text{ gal.}$$

   Valerie knew her answer didn't make sense because there is no way she drinks 20 gallons of water each day. She is not sure where she went wrong.

   a. Discuss Valerie's work with your partner. Then, work together to find the correct number of gallons she drinks each day.

b. Help Valerie understand and learn from her mistakes. Write a brief note to Valerie explaining what her mistakes were and how she can learn from them. Include strategies that may be helpful.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

c. Read your partner's note to Valerie and offer suggestions on ways to improve their note.

5.10.5 Your Turn!

1. Use the conversion rates to find each value. Show your work.

   a. $11{,}200$ milliliters $=$ _____ liters

   b. 3.5 gallons $=$ _____ cups

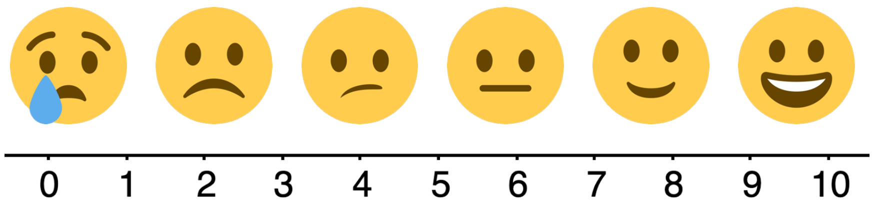
   c. 2 days $=$ _____ minutes

2. Malik plays football for Jones High School in Orlando, Florida. Last Friday, he ran 94 yards for a touchdown on a kickoff return. How many feet did Malik run on the kickoff return?

3. About 727 pints of milk are distributed each day from the cafeteria at Fort King Middle School in Marion County, Florida. How many gallons of milk are distributed at Fort King Middle School each day?

5.10.6 Wrap-Up: Reflection

1. Use the emoji scale to indicate your level of confidence with using ratios to convert units within the same measurement system.



2. Explain why you indicated the confidence level above.

Unit 5, Lesson 11: Solving Problems Involving Ratios

Benchmark

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.

Learning Target

☐ I can solve problems involving ratios, rates, and unit rates.